CSE2315 — Assignment 7

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Exercise 1

Prove that the following problem is in NP:

$$\{(V, E) \mid \text{the graph } G = (V, E) \text{ contains a simple path } \pi \text{ such that } \forall v \in V : v \in \pi\}$$

In a different notation we would write:
Name: Hamiltonian Cycle (HAMC)
Input: A graph $G = (V, E)$.
Question: Is there a simple cycle π in G such that all vertices $v \in V$ are in that cycle?

Proof. The following is a verifier V for $HAMC$:

$V =$ "On input $\langle G, c\rangle$:

1. Test whether c is a tuple containing all nodes in G exactly once.
2. For each i between 1 and $m - 1$ (where m is the number of nodes in G): check that there is an edge in G from node c_i to c_{i+1}
3. Check that there is an edge in G from c_m to c_1

Stages 1. and 3. run exactly once, and stage 2. runs m times, so there is a polynomial amount of stages. We can easily see that each stage can be completed in polynomial time, so the verifier is overall polynomial. Therefore, $HAMC \in NP$ $\square$

Exercise 2

Old exam question, endterm 23–24:

For the following claim, if it is true: prove it. If it is false, give an explanation showing the claim is false and if possible a counterexample. Remember that the cover of our exam reads: "Unless otherwise stated we assume that $P \neq NP$ and that $NP \neq EXP$."

For three languages A, B, C if $A \leq_P B$ and $B \leq_m C$ with $C \in NPC$. Furthermore we know that there is a regular expression R such that $L(R) = A$. Then $B \in P$.

Exercise 3

Old exam question:

On the (famous?) British Quiz Show "The University Challenge", the following question was asked in the grand final of 2018/2019.

...which problem in computing asks whether all problems with a solution verifiable in finite time, can also be solved in finite time.

The contestants answer with “P equals NP” and the host answers back with “Yes that’s right, P versus NP, yes”.

Assuming this is the answer the host was looking for, what is wrong with the question he asks?

Exercise 4

Old exam question, resit 20–21:

(a) Prove that the following problem is in NP.

Name: AmazingCustomChallenge
Instance: Given a set L of lectures, a set of students S, a collection of subsets T with $|T| = |S|$ so that T_i is the set of lectures that student i needs to attend, and an integer K.
Question: Is there a function $f : L \to \{1, 2, \dots, K\}$ that maps lectures to time slots so that all students can attend all their lectures. In other words, is there a function so that there is no student who has to attend two lectures that are mapped to the same time slot.

Proof. The following is a NTM N that decides AmazingCustomChallenge in nondeterministic polynomial time:

$N =$ ”On input $\langle L, S, T, K\rangle$:

1. For each lecture $l \in L$, nondeterministically choose a value $f(l) \in \{1, \dots, K\}$
2. For every student i check that no lectures in $T_i = \{l_{i_1}, \dots, l_{i_x}\}$ are assigned the same time-slot by looking them up for every pair (l_{i_m}, l_{i_n})
3. Accept if no conflict is found, reject otherwise

Stage one has to guess $|L|$ numbers, and writing down each number would take $O(\log K)$, so overall stage 1 takes $O(|L| \log K)$. Stage 2 needs to check all pairs of lectures in every T_i, so this would be bounded by $O(\sum_{i=1}^{|S|} |T_i|^2)$, which would be a worst case of $O(|S| \cdot |L|^2)$. Stage 3 is constant. Therefore, overall machine N runs in $O(|L| \log K + |S| \cdot |L|^2)$, which is polynomial in the size of the input. □

(b) Not originally part of the exam question. Your answer to the previous question should have included time complexities. If we are to describe them in terms of $|I|$ where I is an instance of the problem, what would be the complexity? (Hint: first think about $|I|$, is it $|S|$, or $|L|$, or some combination? How does K factor into this?)

$$|I| \in O(|L| + |S| + \sum_i |T_i| + \log K)$$
$$|I| \in O(|L| + |S| + |S| \cdot |L| + \log K)$$

So, overall the runtime could be expressed roughly as $T(I) \in O(|I|^3)$.

Exercise 5

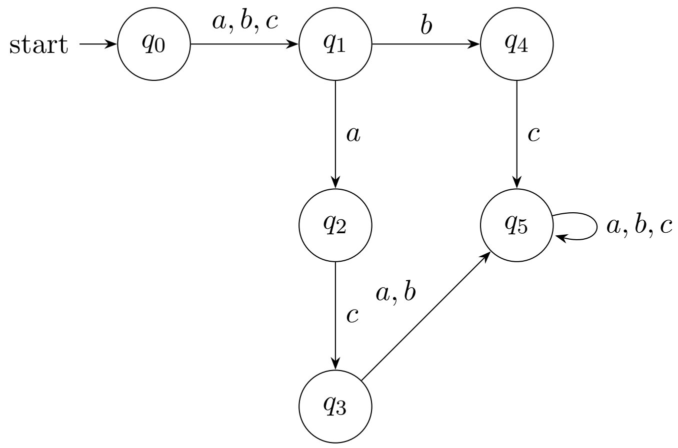
Variation on old exam question, resit 20–21:

Show the claim is true or false, starting your answer with either "True" or "False".

Given two problems A and B that can be solved using a PDA, it is possible to create a Karp-reduction from A to B and vice versa.

Exercise 6

Revision, old exam question, resit 20–21: Given this NFA N, give a regular expression R so that $L(N) = L(R)$.



Exercise 7

Revision, old exam question, endterm 16–17: Suppose we have the following language:

$$L = \{\langle M, v, w\rangle \mid M \text{ outputs } w \text{ on the tape when run on input } v\}$$

Use a mapping reduction to prove that L is undecidable. Give:

(a) A suitable problem to reduce from.

(b) A suitable reduction function $f : \Sigma^* \to \Sigma^*$.

(c) A proof showing f satisfies the requirements of a mapping reduction.

Bonus Exercise